



Differentiating recurrent Guillain–Barre syndrome and acute-onset chronic inflammatory polyneuropathy: literature review

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Abstract

Guillain–Barre syndrome (GBS) is an acute-onset immune-mediated polyneuropathy characterized by ascending symmetrical muscle weakness, diminished reflexes, and sensory symptoms. While GBS typically follows a monophasic course, some patients experience treatment-related fluctuations or recurrences, posing diagnostic challenges in distinguishing GBS from acute-onset chronic inflammatory polyneuropathy (A-CIDP). A-CIDP, may present acutely, simulating GBS, with a nadir in less than 8 weeks, subsequently evolving into a chronic or relapsing course. The distinction between recurrent GBS and A-CIDP is crucial, as A-CIDP necessitates long-term immunosuppression. A PubMed search was conducted using the search terms ‘recurrent Guillain Barre syndrome’ and ‘acute onset CIDP’ focusing on studies in the English language, published between January 1, 2004 and April 30, 2023. Overlapping clinical features, particularly in the early stages, complicate differentiation between recurrent GBS and CIDP. Electrophysiological studies, ultrasonography, and immunological markers have been explored for discrimination; however, definitive criteria for differentiation remain elusive. Recent follow-up studies have further blurred the boundaries between recurrent GBS and A-CIDP, suggesting the persistence of underlying immune processes even in GBS patients without clinical deterioration. This emphasizes the necessity of reevaluating diagnostic criteria and treatment strategies. In conclusion, distinguishing recurrent GBS from A-CIDP remains an ongoing challenge. Existing evidence questions the categorization of recurrent GBS as a distinct entity, challenging its very existence. Continued research is necessary to refine diagnostic criteria and deepen our understanding of these conditions, ultimately advancing patient care. This review delves into the intricacies of recurrent GBS and A-CIDP differentiation and highlights the need for a reevaluation of the recurrent GBS concept.

Keywords Recurrent Guillain–Barre syndrome · Acute-onset chronic inflammatory demyelinating polyneuropathy · Treatment-related fluctuations · Inflammatory polyneuropathy

Introduction

Guillain–Barre syndrome (GBS) is an acute-onset immune-mediated polyneuropathy with an incidence of 1–2/1,00,000 people per year [1–5]. GBS is typically characterized by ascending symmetrical muscle weakness and absent or decreased deep tendon reflexes. Patients may also have

sensory and autonomic symptoms [2]. GBS has several variants, with acute inflammatory demyelinating polyneuropathy (AIDP) constituting approximately 85–90% of all GBS cases in the United States and Europe [6]. In contrast, the primary axonal forms of GBS, acute motor axonal neuropathy (AMAN) and acute motor and sensory axonal neuropathy (AMSAN), are more prevalent in Asia [7]. Miller Fisher syndrome (MFS) and Bickerstaff brainstem encephalitis (BBE) are other common variants [3]. GBS is often triggered by an immune response to a preceding infection or other event that cross-reacts with peripheral nerve components because of molecular mimicry [3]. *Campylobacter jejuni* infection is the most common antecedent event in GBS [8]. Cytomegalovirus (CMV), Influenza A and B, human immunodeficiency virus (HIV), COVID-19 virus, and Zika virus are some of the other viruses which have been associated with GBS

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[8–14]. GBS has also been linked to vaccinations, trauma, and surgery [15, 16].

GBS has a monophasic course and reaches its peak severity within 4 weeks in most cases [1, 2]. However, approximately 5–6% of patients with GBS exhibit treatment-related fluctuations (TRF) [1, 17–19], defined as worsening of at least one grade on the GBS disability scale or by at least five points on the Medical Research Council (MRC) sum score after improvement or stabilization is achieved by immunotherapy within the first 8 weeks after the disease onset, not occurring more than two times [3, 17, 18, 20]. In addition, recurrences have been reported in 2–5% of patients with GBS [21]. Recurrent GBS is defined as the presence of at least two GBS episodes with a nadir within 4 weeks and with an interval of ≥ 4 months if the recovery is incomplete after the first episode or an interval of ≥ 2 months if the recovery is complete / near-complete [22]. On the other hand, chronic inflammatory demyelinating polyneuropathy (CIDP) is a chronic acquired immune-mediated neuropathy [23], typically characterized by clinical worsening over the course of at least 8 weeks [23, 24]. Yet, 16% of patients with CIDP, classified as acute-onset CIDP (A-CIDP), may present acutely with a nadir in less than 8 weeks simulating GBS, followed by a chronic or relapsing course beyond 8 weeks [20, 25, 26]. Patients with three or more TRFs are also included in this definition [25]. However, boundaries between diagnoses are not crystal-clear in all cases. There are overlapping clinical presentations, especially in the early stages, and 2–5% of patients initially diagnosed with GBS turn out to have A-CIDP subsequently [20].

Distinguishing the patients with GBS and A-CIDP correctly is crucial because patients with A-CIDP require long-term immunosuppression. Hence, we aim to provide a narrative review of literature on recurrent GBS (AIDP, AMAN and AMSAN variants) and A-CIDP.

Materials and methods

We conducted a PubMed search using the search terms ‘recurrent Guillain Barre syndrome’ and ‘acute onset CIDP’ to reach all article types written in English language. All relevant articles published between January 1, 2004 and April 30, 2023 were examined.

Results

Studies about recurrent GBS

Recurrent GBS is a contentious diagnosis reported to be present in 2–5% of patients with GBS in the literature [21, 27]. Several studies and case series have provided insights

into the common characteristics and distinctions among recurrent GBS cases. Various studies have indicated that younger patients are more likely to experience recurrences [22, 27–29], particularly those under 30 years of age [29]. However, there have been cases of recurrent GBS presented with first episode after the age of 60 [30–32]. Recurrences are more frequent in AIDP when compared to axonal variants of GBS [22, 27, 33]. Antecedent illness serves an independent risk factor for recurrent GBS [22]. The preceding events may vary [9, 12, 29–31, 34–39] or remain consistent [10, 32, 40–42] in recurrent episodes. Furthermore, a recent study has demonstrated that a high level of thyroid-stimulating hormone (TSH) is another independent risk factor for recurrent GBS [22].

Clinical features during episodes are usually similar with varying severity [29, 32, 36, 39, 43, 44]. Some cases have shown milder symptoms in recurrent episodes compared to the initial episode [12, 37, 41], while others have experienced exacerbations of symptoms or more severe presentations in following episodes [9, 30, 31, 40, 45, 46]. The time interval between recurrent episodes varies considerably in the literature. In addition to cases of recurrence after as short as 2 months, cases with long periods of up to 43 years between attacks have also been reported [34, 35, 37, 41–43, 46]. Moreover, previous studies have shown that time from onset to nadir of the recurrent episodes slightly diminishes during the course, and time from the triggering infection to the onset of recurrence tends to shorten in successive episodes [31, 33, 34].

The presence of anti-ganglioside antibodies (i.e. GM1, GM2, GM3, GM4, GD1a, GD1b, GD2, GD3, GT1a, GT1b, GQ1b, GalNAc-GD1a), especially IgG type, is known to be pathogenic in GBS patients [35, 38, 42], while the role of anti-ganglioside IgM antibodies is still a subject of debate [43]. Moreover, the pattern of antibody positivity may differ from one episode to another in the same patient. The positivity of same [32, 36] or distinct [12, 40] anti-ganglioside antibodies in the recurrent episodes have been documented.

The literature also presents a limited number of cases with recurrent axonal GBS [30, 32, 33, 36]. In a retrospective study from Sweden, one of the fifteen recurrent GBS cases was the axonal variant [33]. In another case report, a 61-year-old male patient with a history of AMAN experiencing a recurrence following the second dose of the shingles vaccine has been reported [30]. Jo et al. reported a 75-year-old female who underwent two episodes of AMAN after a urinary tract infection, 8 months apart. Anti-GM1 and anti-GD1a IgG antibodies were positive in serum in both episodes [32]. Another case was a 37-year-old male with three episodes of acute axonal motor neuropathy. First and second episodes were 54 days apart, with full recovery, both were following diarrhea. The third episode was 3 years later. Anti-GM1, anti-GM2, anti-GD1b, and anti-GalNAc-GD1a

antibodies were positive in the first and third episodes. The authors initially interpreted the case as TRF after the second episode and later diagnosed it as recurrent GBS [36].

Recurrent GBS may also manifest as a part of combined central and peripheral demyelination (CCPD). We had a young male patient with recurrent simultaneous central and peripheral nervous system demyelination, 12 years apart. The peripheral nervous system involvement was compatible with two episodes of recurrent GBS. Anti-neurofascin-155 and anti-neurofascin-186 antibodies were negative. The prognosis was excellent with pulse methylprednisolone, intravenous immunoglobulin, and plasmapheresis [47].

Studies about A-CIDP

A-CIDP represents a unique subset of CIDP with acute onset of symptoms followed by a chronic course lasting beyond 8 weeks and associations with various antecedent events. Herein, we have examined the literature, encompassing studies and case reports, to present the properties and clinical characteristics of A-CIDP patients.

In CIDP, the frequency of antecedent events, including infections and vaccinations, has been reported to range between 10 and 33% [48]. On the other hand, chronic inflammatory diseases such as systemic lupus erythematosus and sarcoidosis are predominantly associated with CIDP rather than GBS [49–52]. Notably, CIDP cases associated with antecedent events are more likely to be classified as A-CIDP [48, 53], occur at a younger age [54], and exhibit cranial nerve involvement [48, 54, 55] than those without antecedent events. Recently, A-CIDP cases following COVID-19 infection or vaccination have been reported [11, 13, 56]. Some reported significant clinical improvement over a 2-month follow-up period [11].

Anti-ganglioside antibody testing in chronic immune-mediated neuropathies is not as common as in acute immune-mediated neuropathies, and IgM type anti-ganglioside antibodies are more common in chronic immune-mediated neuropathies than acute ones [43, 57]. An atypical case of A-CIDP has been reported with three episodes in a 3-year period [58]. The first episode was 2 weeks after an upper respiratory infection, presenting with progressive paraparesis. Electroneuromyography (ENMG) revealed prolonged F-wave latencies and the absence of sensory nerve action potentials, suggesting a demyelinating pathology. Recovery was complete after the first episode. The second episode was 1 year later, following a diarrheal illness, presenting with recurrence of paraparesis. ENMG was not performed, and the patient was treated with supportive care. Recovery was incomplete, reaching approximately 80% of the patient's baseline. The last episode occurred 3 years after the initial episode, with worsening quadriparesis. ENMG revealed pure axonal features which has been interpreted as

axonal loss due to progressive untreated demyelination or primary axonal neuropathy by the authors. Anti-GM1, anti-GD1a, and anti-GD1b antibodies have been positive which are known to be prominent in AMAN but have also been seen in CIDP variants. Eight months later, the recovery was partial [58]. Other intriguing cases include A-CIDP evolving from post-partum GBS [59] and VEXAS (vacuoles, E1 enzyme, X-linked, autoinflammation, somatic) syndrome [60], Zika virus-associated A-CIDP [14], A-CIDP following Hantavirus infection in a hepatitis B virus carrier [61], and after Influenza vaccination [26].

Over the past few decades, several antibodies (anti-contactin-1, anti-neurofascin-155, anti-neurofascin-186, anti-contactin-associated protein-1) against nodal/paranodal proteins have been identified in a subset of patients with peripheral neuropathies [62]. Initially classified as CIDP according to the European Federation of Neurological Sciences / Peripheral Nerve Society (PNS) 2010 guideline [24], these cases have recently been redefined as a distinct category known as autoimmune nodopathies. The European Academy of Neurology / PNS 2021 guideline [63] now recognizes them as separate entities, distinct from CIDP, due to their specific clinical features and responses to treatment. It is imperative to bear in mind that certain patients with clinical presentations resembling A-CIDP may indeed be afflicted by autoimmune nodopathies. This is particularly relevant for anti-neurofascin-186 antibody positive ones, as they can present symptoms indicative of acute or subacute onset acquired demyelinating polyneuropathies [64].

Studies about differentiating A-CIDP from GBS

The differentiation between A-CIDP and GBS with or without recurrences poses a significant clinical challenge due to their overlapping clinical features and distinct treatment strategies. Moreover, in some instances, CIDP may have a monophasic course and enter complete remission [65], making the differential diagnosis from GBS more challenging. Some clinical clues have been suggested to distinguish between GBS and A-CIDP [20, 25, 66]. Recently, several ultrasonographic and immunological markers have also been proposed [67–70]. Nonetheless, how to differentiate A-CIDP from GBS electrophysiologically remains unclear [66].

Patients with A-CIDP are more likely to have an older age at disease onset [25], exhibit prominent sensory symptoms, impaired vibration sensation, sensory ataxia [66] and less likely to have cranial, and autonomic involvement, and loss of pinprick sensation compared to patients with GBS [20, 25, 66]. Additionally, a history of type 2 diabetes mellitus is more frequent in patients with A-CIDP [25]. On the other hand, there are discrepancies in the literature regarding the need for mechanical ventilation in patients with A-CIDP and GBS. While one study has been found similar rates for

mechanical ventilation in both patient groups [25], other studies have been stated that the need for mechanical ventilation is more common in patients with GBS [20, 66]. In addition, patients with GBS can be differentiated from CIDP by long asymptomatic periods, rapid onset of symptoms, normal cerebrospinal fluid (CSF) proteins in the early phase, and more frequent antecedent infections [29, 66]. However, in a study comparing patients with A-CIDP and fluctuating GBS, the frequency of antecedent infections has been similar in both groups of patients [20]. Another distinction between these diseases is the time from neurological symptoms to maximum disability, which is often less than 4 weeks in patients with GBS, while by definition, it is more than 8 weeks in patients with CIDP [24].

Distinguishing between A-CIDP and AIDP through electrophysiological means remains challenging. Currently, no specific electrophysiological markers, including sural-sparing pattern, sensory ratio, or presence of A waves have been identified to definitely distinguish between these two neuropathies [66]. There are conflicting results regarding sural-sparing pattern and the presence of A waves. While one study indicates that these parameters are more commonly seen in AIDP [68], another study suggests that these parameters do not differ significantly in A-CIDP and AIDP [66]. Nevertheless, in a prospective study, they speculated that nerve excitability studies may assist in differentiating A-CIDP from AIDP in the early stages. They found abnormalities in the recovery cycle of excitability, including significantly reduced superexcitability and prolonged relative refractory period without changes in threshold electrotonus in AIDP. On the other hand, different pattern changes observed in the recovery cycle, suggesting early repolarization in A-CIDP [71]. Therefore, it is difficult to differentiate A-CIDP from AIDP electrophysiologically in the early stages, and the usefulness of electromyographic follow-up in differentiation of those pathologies is also debatable. It has been stated that as the disease progresses, the sural nerve might become affected in CIDP and sural-sparing pattern might be used for differential diagnosis [66]. On the other hand, in a recent follow-up study of patients with AIDP, it was reported that approximately 39% of patients continued to meet the definite CIDP criteria 1 year after the acute AIDP episode, and approximately 10% of the patients did so more than 18 months after. Conduction blocks were present in three patients on nerve conduction studies at 1 year, suggesting active demyelination [72]. So, nerve conduction study results should be interpreted cautiously with symptoms and clinical examination findings during follow-up, before reaching a diagnosis.

Ultrasonography is another modality that has been studied for A-CIDP and GBS discrimination. The Bochum ultrasound score (BUS), based on cross-sectional areas of ulnar, radial, and sural nerves, has emerged as a potentially

valuable tool. Given the higher Bochum scores indicating increased thickness of the nerve; presence of sensory symptoms and a sum BUS of ≥ 2 points were more frequent in A-CIDP and had the highest sensitivity and specificity in distinguishing A-CIDP from GBS [67, 73]. Similarly, in a recent study they demonstrated that patients with CIDP had significantly enlarged nerves, except for the roots and vagus, which were enlarged in both CIDP and GBS. Moreover, pure sensory nerves were mostly enlarged in CIDP, while remaining normal in GBS. This ultrasonic sensory-sparing pattern combined with enlarged roots / vagus has been defined as a hallmark of GBS [68].

Immunological differences between A-CIDP and GBS have also been explored. Studies have indicated that Fas-mediated T-cell apoptosis is impaired in A-CIDP but not in GBS. This suggests that evaluating Fas function may aid in the early identification of GBS and A-CIDP [70, 74]. Additionally, CSF markers have also been investigated. In a retrospective study, they have found higher CSF interleukin-8 (IL-8) levels in patients with GBS compared to those with CIDP, and proposed an optimal cut-off value of 70 pg/mL [69].

The differential features between A-CIDP and GBS are presented in Table 1.

Discussion

The differential diagnosis between A-CIDP and GBS, or whether recurrent GBS is a real entity or not, remains the subject of ongoing debate and poses several challenges. While there is some consensus on distinguishing these two entities, it is crucial to critically analyze existing studies to question the existence of recurrent GBS. The traditional definition of recurrent GBS includes episodes with a time to nadir of no more than 4 weeks [33]. However, it is worth noting that some authors include relapsing–remitting cases into the CIDP category, such as those with intermittent episodes associated with respiratory failure [75] or recurrence after 18 months of full recovery [66]. CIDP progresses gradually over more than 8 weeks. However, patients with A-CIDP present with more rapidly progressive symptoms. Some authors have described A-CIDP with a time to nadir of less than 4 weeks [20]. GBS is often caused by an immune response to a preceding infection, or other event that cross-reacts with peripheral nerve components [3, 8]. Although the exact cause of CIDP is unknown, CIDP is also an immunologically mediated neuropathy with multiple possible triggers. Both cellular and humoral immunity are involved in the pathogenesis of CIDP [48–51]. Although the pathophysiology of CIDP and GBS is quite different, there is not enough data whether recurrent GBS is a unique entity or another

Table 1 The differential features between acute-onset chronic inflammatory polyneuropathy and Guillain–Barre syndrome

Author	Parameter	A-CIDP	GBS	<i>p</i> value
Dionne et al.	Impaired proprioception; <i>n</i> (%)	7 (46.7)	4 (13.3)	0.030
Dionne et al.	Sensory ataxia; <i>n</i> (%)	8 (53.3)	1 (3.0)	<0.001
	Cranial nerve involvement			
Alessandro et al.	<i>n</i> (%)	5 (36)	36 (47)	NS
Ruts et al.	<i>n</i> (%)	1 (13)	11 (69)	0.030
	Autonomic dysfunction			
Alessandro et al.	<i>n</i> (%)	6 (46)	28 (36)	NS
Dionne et al.	<i>n</i> (%)	2 (13.3)	21 (70)	<0.001
	Need for MV			
Alessandro et al.	<i>n</i> (%)	1 (7)	5 (6)	NS
Dionne et al.	<i>n</i> (%)	1 (6.7)	11 (36.7)	0.040
Ruts et al.	<i>n</i> (%)	0 (0)	7 (44)	0.050
Alessandro et al.	Type 2 DM; <i>n</i> (%)	4 (29)	6 (8)	0.040
	Antecedent infections			
Ruts et al.	<i>n</i> (%)	4 (50.0)	9 (56.3)	NS
Dionne et al.	<i>n</i> (%)	5 (33.3)	23 (76.7)	0.008
	Sural-sparing pattern			
Dionne et al.	<i>n</i> (%)	7 (46.7)	20 (66.7)	0.22
Grimm et al.	<i>n</i> (%)	17 (50)	24 (73)	0.048
Dionne et al.	Sensory ratio > 1; <i>n</i> (%)	5 (41.7)	17 (63.0)	0.3
	Presence of A waves			
Dionne et al.	<i>n</i> (%)	4 (40.0)	11 (52.4)	0.7
Grimm et al.	<i>n</i> (%)	6 (18)	19 (58)	0.001
Kerasnoudis et al.	Bochum ultrasound score ≥ 2 points; <i>n</i> (%)	9 (90)	3 (20)	<0.001
Grimm et al.	Nerve enlargement	Significant nerve enlargement, except for roots and vagus	Enlarged roots and vagus	<0.001
Grimm et al.	u-Sensory nerve sparing	Rare	Common	<0.050
Breville et al.	CSF IL-8 level; median (IQR)	41.0 pg/ml (17.3)	83.9 pg/ml (64.4)	<0.001
Comi et al.	Fas-mediated T-cell apoptosis	Defective	Normal	NA

Bold indicates statistical significance

A-CIDP acute-onset chronic inflammatory demyelinating polyneuropathy, *GBS* Guillain–Barre syndrome, *MV* mechanical ventilation, *DM* diabetes mellitus, *u* ultrasonic, *CSF* cerebrospinal fluid, *IL* interleukin, *IQR* interquartile range, *NS* not significant, *NA* not available

variant of CIDP. Some authors speculate that instead of being distinct entities, recurrent GBS and CIDP may represent a spectrum of inflammatory demyelinating polyradiculopathies [65, 76]. Drawing an analogy with the differentiation between clinical isolated syndrome (CIS) and multiple sclerosis (MS) can further support the idea that recurrent GBS may actually be part of the CIDP spectrum. While CIS refers to a single clinical demyelination episode resembling GBS, CIDP shares similarities with the chronic and relapsing nature of MS. On the other hand, differentiating A-CIDP from GBS in the early stages is challenging but crucial as they require different treatment strategies. Unfortunately, despite some clinical clues [20, 25, 66], ultrasonographic and immunological markers [67–70], and electrophysiological studies [66, 68] differentiation of A-CIDP from GBS remains elusive.

In a retrospective study, it was showed that nerve conduction studies continued to worsen up to 18 months after the onset of symptoms in AIDP, and CIDP-like abnormalities suggestive of demyelination were identified [72]. These findings suggest that the underlying autoimmune process may persist in a subset of AIDP patients, even in the absence of clinical deterioration. The findings may also potentially explain residual symptoms such as paresthesia, severe pain, or fatigue [77] that are observed in some individuals after a single acute GBS attack. Based on this study, it is important to reconsider treatment strategies for patients with recurrent GBS and A-CIDP. Patients classified as having recurrent GBS after a second attack do not receive long-term immunotherapy, while those diagnosed with A-CIDP do receive. However, we believe that this approach may potentially increase the disabilities of patients with recurrent

GBS due to the underlying, subtly ongoing immune process. Therefore, it is crucial to closely monitor patients after recurrent episodes, evaluate the initiation of immunotherapy, and even reevaluate the diagnosis. By doing so, we can provide appropriate treatment strategies and potentially improve outcomes.

Reconsidering certain aspects, including cases that are positive for antiganglioside antibodies, cases with a relapsing axonal presentation (AMAN, AMSAN), and cases with antecedent events, is essential when suggesting that patients with recurrent GBS might be acute-onset axonal or demyelinating CIDP cases. Although less common than in GBS, anti-ganglioside antibody positivity and antecedent events such as vaccinations and infections can also be observed in patients with CIDP [35, 38, 42, 48, 57]. In recent years, the term of “chronic inflammatory axonal polyneuropathy (CIAP)” has been proposed instead of axonal CIDP. It has been stated that CIAP, by definition, progresses over a period of > 8 weeks, and the distinction between CIAP and axonal CIDP lies in the assumption that in axonal CIDP, axonal changes are secondary to CIDP. Moreover, among the patients presented, there are two cases that initially presented as axonal GBS but were later diagnosed with CIAP / axonal CIDP [78], suggesting the presence of acute-onset axonal CIDP cases.

Conclusion

The differentiation between A-CIDP and GBS, especially recurrent cases, presents a significant challenge within the field of neurology. Existing evidence raises doubts regarding the distinct classification of recurrent GBS as a separate clinical entity. We argue that the arbitrary distinction between recurrent GBS and CIDP, the overlapping clinical presentations, and the presence of ongoing autoimmune processes all underscore the need for a reevaluation of diagnostic criteria for CIDP and recurrent GBS. Moreover, the presence of axonal cases further complicates the clinical landscape. Recurrent GBS may represent another variant of CIDP that has not been fully described yet. Therefore, additional research and clinical investigations are warranted to enhance our understanding of these conditions, ultimately leading to the development of more effective treatment strategies and advanced patient care.

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